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SE-AIML

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Expt.-7: Ensemble Learning for House Price Prediction

In [1]:
```
import time
import pandas as pd
from sklearn.datasets import fetch_california_housing
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
```

In [2]:
```
data = fetch_california_housing()
print(data.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'feature_names', 'DESCR'])
```

In [3]:
```
x = pd.DataFrame(data.data, columns = data.feature_names)
y = data.target

print(x.head())
print("Sample:", len(x))
print(y[:10])
```

```
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude
0    -122.23
1    -122.22
2    -122.24
3    -122.25
4    -122.25
Sample: 20640
[4.526 3.585 3.521 3.413 3.422 2.697 2.992 2.414 2.267 2.611]
```

In [4]:
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

In [5]:
```
tree = DecisionTreeRegressor()

start = time.perf_counter()
tree.fit(x_train, y_train)
tree_time = time.perf_counter() - start

pred_tree_test = tree.predict(x_test)
err_tree_test = mean_absolute_error(y_test, pred_tree_test)
pred_tree_train = tree.predict(x_train)
err_tree_train = mean_absolute_error(y_train, pred_tree_train)

print("Decision Tree Train MAE:", err_tree_train)
print("Decision Tree Test MAE:", err_tree_test)
print("Decision Tree Train Time:", tree_time)
```

```
Decision Tree Train MAE: 4.222505205090833e-17
Decision Tree Test MAE: 0.4574647940891473
Decision Tree Train Time: 0.1375576999999999
```

In [6]:
```
rf = RandomForestRegressor(n_estimators=100)

start = time.perf_counter()
rf.fit(x_train, y_train)
rf_time = time.perf_counter() - start


pred_rf_test = rf.predict(x_test)
err_rf_test = mean_absolute_error(y_test, pred_rf_test)
pred_rf_train = rf.predict(x_train)
err_rf_train = mean_absolute_error(y_train, pred_rf_train)

print(" Random Forest Train MAE:", err_rf_train)
print(" Random Forest Test MAE:", err_rf_test)
print(" Random Forest Train Time:", rf_time)
```

```
Random Forest Train MAE: 0.12232079781976782
Random Forest Test MAE: 0.32743024021317846
Random Forest Train Time: 13.3623328
```

In [7]:
```
gb =  GradientBoostingRegressor(n_estimators=300, learning_rate=0.1, max_depth=5)

start = time.perf_counter()
gb.fit(x_train, y_train)
gb_time = time.perf_counter() - start


pred_gb_test = gb.predict(x_test)
err_gb_test = mean_absolute_error(y_test, pred_gb_test)
pred_gb_train = gb.predict(x_train)
err_gb_train = mean_absolute_error(y_train, pred_gb_train)

print(" Gradient Boosting Train MAE:", err_gb_train)
print(" Gradient Boosting Test MAE:", err_gb_test)
print(" Gradient Boosting Train Time:", gb_time)
```

```
Gradient Boosting Train MAE: 0.22968324793032216
Gradient Boosting Test MAE: 0.306392811460534
Gradient Boosting Train Time: 22.770746099999997
```

In [8]:
```
print("\n ==== Comparison For  Test MAE ==== \n")
print("Decision Tree Test:", err_tree_test)
print("Random Forest Test:", err_rf_test)
print("Gradient Boosting Test:", err_gb_test)
```

```
==== Comparison For  Test MAE ====

Decision Tree Test: 0.4574647940891473
Random Forest Test: 0.32743024021317846
Gradient Boosting Test: 0.306392811460534
```

In [9]:
```
print("\n ==== Comparison For Train MAE ==== \n")
print("Decision Tree Train MAE:", err_tree_train)
print("Random Forest Train MAE:", err_rf_train)
print("Gradient Boosting Train MAE:", err_gb_train)
```

```
==== Comparison For Train MAE ====

Decision Tree Train MAE: 4.222505205090833e-17
Random Forest Train MAE: 0.12232079781976782
Gradient Boosting Train MAE: 0.22968324793032216
```

In [10]:
```
print("\n ==== Comparison For Time MAE ==== \n")
print("Decision Tree Train Time:", tree_time)
print("Random Forest Train Time:", rf_time)
print("Gradient Boosting Train Time:", gb_time)
```

```
==== Comparison For Time MAE ====

Decision Tree Train Time: 0.1375576999999999
Random Forest Train Time: 13.3623328
Gradient Boosting Train Time: 22.770746099999997
```